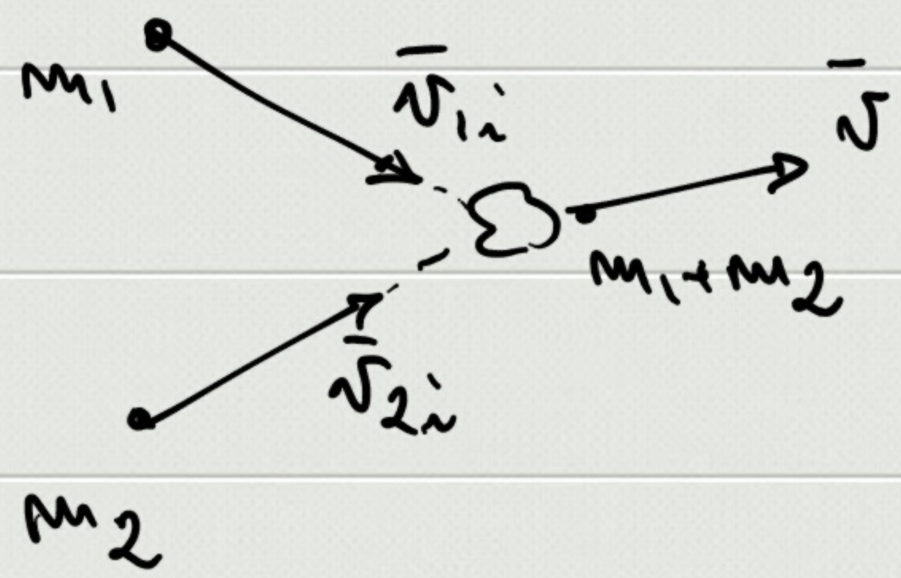
$\boxed{\bar{P} = \text{cost}}$ $\quad E_k \neq \text{cost}$

urto <u>completamente</u> anelastico

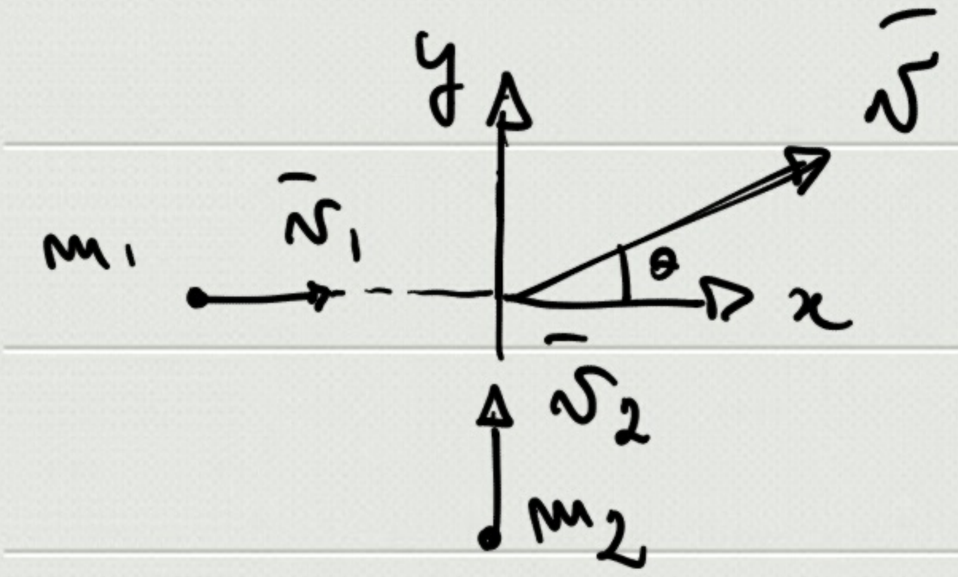


$$\bar{P} = m_1 \bar{v}_{1i} + m_2 \bar{v}_{2i} = (m_1 + m_2)\bar{v} \qquad *$$

$$\overset{!}{=} m_{TOT}\, \bar{v}_{CM} \qquad\qquad m_{TOT} \quad \bar{v}_{CM}$$

$$\Delta \bar{p}_1 = m_1 \bar{v} - m_1 \bar{v}_{1i} \overset{(*)}{\overset{!}{=}} -\left(m_2 \bar{v} - m_2 \bar{v}_{2i}\right) = -\Delta \bar{p}_2$$

$$\Delta E_k = E_{k,f} - E_{k,i} = \frac{1}{2}(m_1 + m_2)\, v^2 + - \left(\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2\right)$$

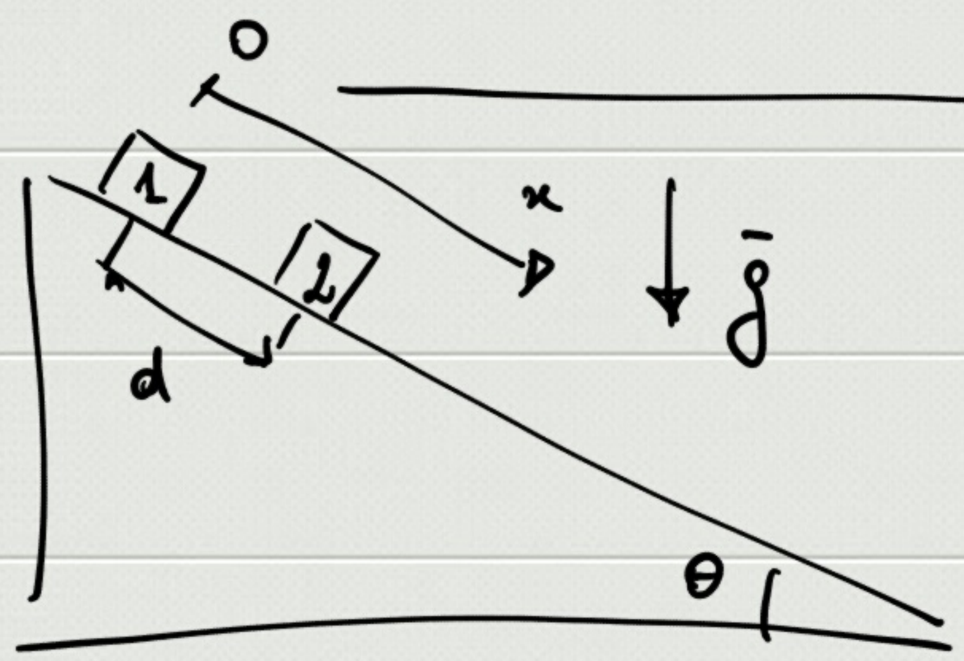


$$m_1 \bar{v}_1 + m_2 \bar{v}_2 = (m_1 + m_2)\, \bar{v}$$

$$\begin{cases} x: & m_1 v_1 = (m_1 + m_2)\, v \cos\theta \\ y: & m_2 v_2 = (m_1 + m_2)\, v \sin\theta \end{cases}$$

$$\operatorname{tg}\theta = \frac{m_2 v_2}{m_1 v_1} \quad \Rightarrow \quad m_1^2 v_1^2 + m_2^2 v_2^2 = (m_1 + m_2)^2 v^2$$

$$\Rightarrow v = \frac{\sqrt{m_1^2 v_1^2 + m_2^2 v_2^2}}{m_1 + m_2}$$



$v_{01} > v_{02}$

urto compl. anel

$v = ?$

$$\bar{P}_i = \bar{P}_f$$

$$\Rightarrow \bar{P}(t^{*-}) = \bar{P}(t^{*+})$$

$$a = g \sin\theta$$

$$1: \; x_1(t) = v_{01} t + \frac{1}{2} a t^2$$

$$2: \; x_2(t) = d + v_{02} t + \frac{1}{2} a t^2$$

$$x_1(t^*) = x_2(t^*) \quad \Rightarrow \quad v_{01} t^* + \cancel{\frac{1}{2} a t^{*2}} = d + v_{02} t^* + \cancel{\frac{1}{2} a t^{*2}}$$

$$t^* = \frac{d}{v_{01} - v_{02}}$$

$\vec{P} = \text{costante} \Rightarrow \qquad m_1 v_1(t^*) + m_2 v_2(t^*) = (m_1 + m_2) v^*$

$v_1(t) = v_{01} + at$

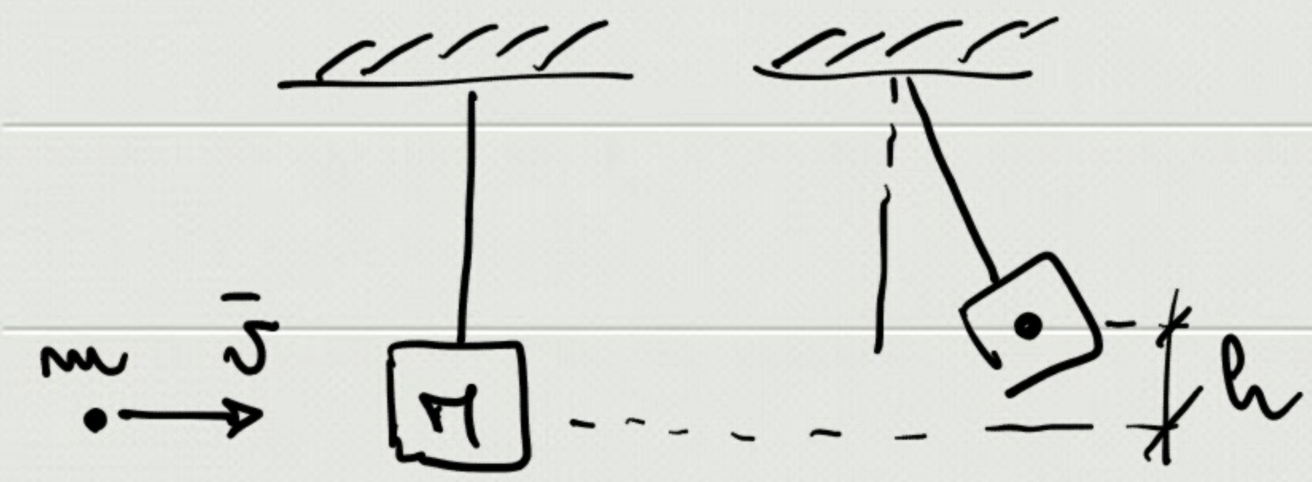
$v_2(t) = v_{02} + at$

$$\Rightarrow m_1 v_{01} + m_1 a t^* + m_2 v_{02} + m_2 a t^* = (m_1 + m_2) v^*$$

$$\Rightarrow v^* = \frac{m v_{01} + m_2 v_{02}}{m_1 + m_2} + a t^*$$

$$\overset{!}{=} v_{CM,0} + a t^*$$

Pendolo balistico



$$mv = (m+M) V \Rightarrow V = \frac{m}{m+M} v$$

$$\frac{1}{2} \cancel{(m+M)} V^2 = \cancel{(m+M)} gh \Rightarrow \frac{m^2}{(m+M)^2} v^2 = 2gh$$

$$\boxed{v = \frac{m+M}{m} \sqrt{2gh}}$$